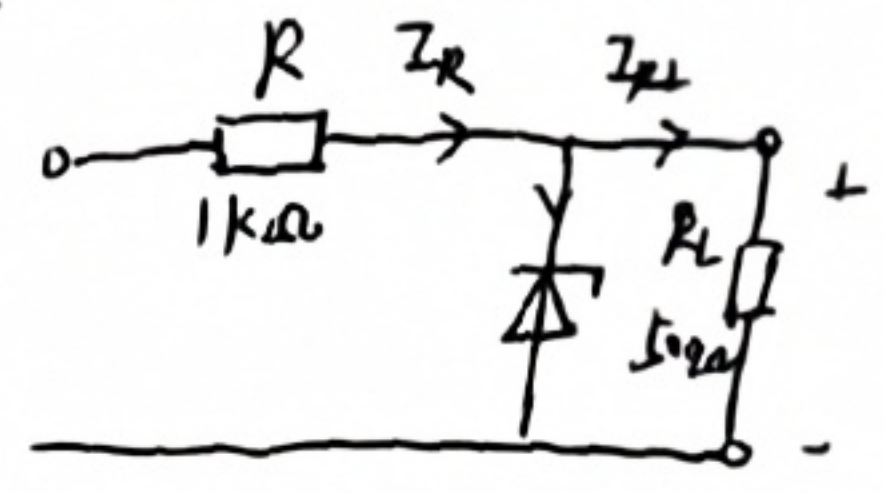


作业 ⑧.

2. 稳压管 $U_Z = 6V$. 最小稳定电流 $I_{Zmin} = 5mA$. 最大: $I_{Zmax} = 25mA$.

(1). $U_1 = 10V, 15V, 35V$ 时, U_0 的值.

(2) $U_1 = 35V$ 时, 负载开路, 会出现什么现象, 为什么?



① $U_1 = 10V$

假设稳压管稳压状态, 则 $U_0 = U_Z = 6V$.

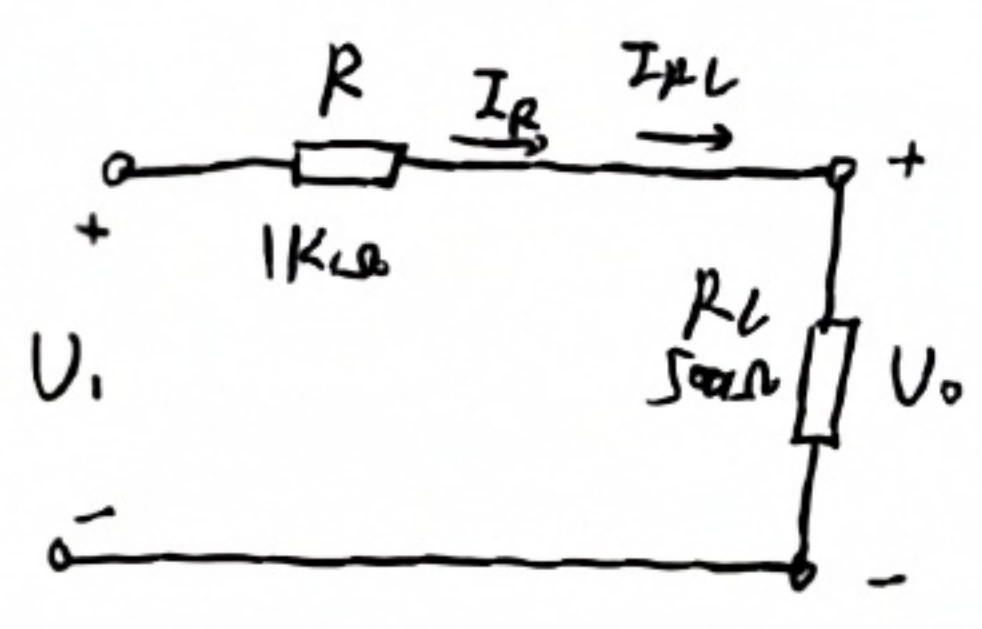
流过电阻 R 电流:

$$I_R = \frac{U_1 - U_0}{R} = \frac{10 - 6}{1000} = 4mA$$

$$稳压管 I_{DZ} = I_R - I_{RL} < I_R$$

即 $I_{DZ} < I_Z < I_{Zmin}$. 即稳压管处于截止状态.

即
电路为



$$U_0 = \frac{R_L}{R + R_L} U_1 \approx 3.33V$$

② $U_1 = 15V$ 时.

假设稳压: $U_0 = U_Z = 6V$.

R :

$$I_R = \frac{U_1 - U_0}{R} = \frac{15 - 6}{1000} = 9mA$$

R_L :

$$I_{RL} = \frac{U_0}{R_L} = 12mA$$

$$I_{DZ} = I_R - I_{RL} = -3mA < I_{Zmin}$$

截止状态:

$$U_0 = \frac{R_L}{R + R_L} \cdot U_1 = \frac{500}{1000 + 500} \cdot 15 = 5V$$

(3) 假设: 稳压.

$$U_0 = U_Z = 6V$$

$$R: I_R = \frac{U_1 - U_0}{R} = \frac{35 - 6}{1000} = 29mA$$

负载开路

$$I_{DZ} = I_R = 29mA > I_{Zmax} = 25mA$$

功耗过大而损坏.

③ $U_1 = 35V$ 时.

假设稳压: $U_0 = U_Z = 6V$

$$R: I_R = \frac{U_1 - U_0}{R} = \frac{35 - 6}{1000} = 29mA$$

R_L :

$$I_{RL} = \frac{U_0}{R_L} = 12mA$$

$$I_{DZ} = I_R - I_{RL} = 29mA - 12mA = 17mA \in [I_{Zmin}, I_{Zmax}]$$

稳压状态. 即 $U_0 = U_Z = 6V$.